Erasmus University Rotterdam
Erasmus School of Economics
Bachelor Thesis (FEB63007H) - BSc Econometrics and Operations Research

Option-driven improvements in point and density forecasting of volatility

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Date final version:	2nd March 2026

Abstract

This paper studies the gains of incorporating option-based measures in point and density forecasting of volatility. Using short-dated S&P 500 index options, we construct a nonparametric estimator of spot volatility. Combining the option-based estimator with a measure based on high-frequency returns leads to more precise measurement of volatility. We empirically show that this, in turn, leads to substantial gains in volatility point forecasting. Using shrinkage techniques, we show that parsimonious model specifications using only an option-based and return-based volatility measure as predictors are effective in forecasting. An empirical analysis reveals that relaxing the distributional assumptions of forecasting errors leads to a more accurate representation of forecast uncertainty. Under these flexible assumptions, we find evidence that the gains of adding option data also extend to density forecasting.

The views stated in this thesis are those of the author and not necessarily those of the supervisor, second assessor, Erasmus School of Economics or Erasmus University Rotterdam.

1 Introduction

Volatility is a measure of variability in the returns of a financial asset. Accurate volatility forecasts are crucial in sectors such as portfolio management, asset pricing, asset allocation and risk management. For instance, they allow asset managers to make better risk-return trade-offs when adjusting their portfolios. An inherent problem to forecasting volatility is that volatility itself is unobserved. Therefore, volatility is instead inferred through different econometric models.

Generalized autoregressive conditional heteroscedasticity (GARCH) models have traditionally been used for volatility forecasting (Bollerslev, 1986). However, the increased availability of high-frequency financial data has led to an increased focus on nonparametric estimates of volatility. The most prominent measure is realized volatility (RV), defined as the sum of squared intra-day returns (Andersen and Bollerslev, 1998). Given enough intra-day observations, volatility essentially becomes an observable variable, which can be directly modeled and forecasted through time series. Many other return-based nonparametric volatility estimates have been proposed (see, e.g., Barndorff-Nielsen and Shephard (2004); Barndorff-Nielsen and Shephard (2006); Jacod and Todorov (2014); Mancini (2009)). It is now generally accepted that models based on nonparametric volatility measures outperform traditional GARCH-type models (Bekaert and Hoerova, 2014).

Parallel to the increasing availability of high-frequency return data, there has been a rapid increase in option trading (Andersen et al., 2017). As the volatility of an underlying asset is directly related to the likelihood of an option ending in-the-money, option prices naturally reflect expectations about current and future volatility. Nevertheless, it is not immediately clear how a measure of volatility can be extracted from option data. Using the increased availability of options with low maturity, Todorov (2019) addresses this by developing an estimator that nonparametrically estimates the true underlying spot volatility of an asset. Notably, the measurement error of this option-based estimator is driven by noise in the option prices at the time of observation, whereas the measurement error of return-based estimators arises from stochastic variability in an asset's price process over the observation window. As the errors of these estimates arise from different sources, the measurement errors of the return-based and option-based estimators should be asymptotically uncorrelated.

Ghysels et al. (2023) show that if the errors of two alternative volatility estimators are asymptotically uncorrelated, one can improve measurement of volatility by combining the two estimators. In other words, combining alternative estimates of volatility allows us, at least theoretically, to 'diversify' their measurement errors and thereby improve precision in measurement, which should consequently lead to gains in forecasting. To assess this theoretical concept in practice, our paper aims to empirically quantify the forecasting gains of combining the option-based and return-based volatility measures in an application to the S&P 500 index.

However accurate volatility forecast may be, volatility itself is still unobservable, and forecasts will always carry uncertainty. Quantifying the uncertainty around a volatility forecast provides a better expectation of how much the true volatility of an asset might deviate from a volatility forecast. Alongside benefits in other applications, adjusting for this uncertainty can in turn help improve the precision of return forecasts (Corsi et al., 2008). To quantify the variability around point forecasts, we construct density forecasts by explicitly modeling the distribution

of forecast errors. We then investigate whether combining the option-based and return-based volatility measures can also help improve density forecasts. In summary, our study thus aims to answer the following research question:

> *What is the incremental value of option-augmented models over return-only models in nonparametric volatility forecasting, and do these gains extend to density forecasting?*

To answer this research question, we calculate multiple nonparametric high-frequency return and option-based measures for the volatility of the S&P 500 index. We first derive the realized volatility (RV), and truncated volatility (TV) of Mancini (2009), using high-frequency intra-day return data. Then, we construct an option-based volatility estimator (OV) proposed by Todorov (2019). The estimator nonparametrically recovers the risk-neutral distribution of price increments through a set of out-of-the-money (OTM) options written on the underlying asset. Furthermore, following Todorov (2019), we isolate and estimate a risk-neutral jump component (JV) derived from OTM options, allowing a discontinuous jump process to be separated from a continuous volatility path. Finally, we introduce a more precise measure of volatility (EV) by optimally combining return-based and option-based volatility proxies based on asymptotic variance, thereby 'diversifying' their respective measurement errors.

To assess the forecasting gains of adding an option-based volatility measure to a high-frequency return-based measure, we perform rolling and expanding window forecasts of the S&P 500 index volatility at daily, weekly, and monthly horizons. The heterogeneous autoregressive (HAR) model introduced by Corsi (2009), originally based on high-frequency return measures, is compared and extended with the option-implied volatility measures introduced in this study. Our results reveal that the measurement gains of combining option-based and return-based volatility estimators also translate into significant forecasting gains. At the daily forecasting horizon, the time series median of the forecasting loss is reduced by up to 30%. While the gains at the weekly and monthly forecasting horizons are slightly smaller, they remain substantial. Finally, we show that the gains of adding option data become even more substantial when forecasting a less noisy future volatility proxy.

Building upon this analysis, we extend the forecasting framework by integrating daily, weekly and monthly lags of all introduced option-implied and return-based volatility measures into a single general forecasting model, following the approach of Zhang et al. (2019). In addition to the return-based measures RV and TV, as predictors for future volatility we incorporate the option-based spot volatility measure OV, the option-based jump component JV, the optimally combined measure EV, and the forward-looking VIX index. First of all, as each of these volatility measures has a time-varying measurement error stemming from different sources, their predictive ability may also vary over time. Second, incorporating the VIX index allows capturing market expectations about future volatility that may have not been fully reflected in our constructed volatility measures. To assess whether combining all measures improves forecasting performance, we evaluate shrinkage methods that adaptively select the most relevant predictors at each point in time. Our results reveal that the benefit of adaptive selection is limited. In particular, forecasting gains beyond a HAR specification combining the return-based TV and option-based OV are relatively small. Additional gains of incorporating the VIX index as predictor also remain limited, illustrating the effectiveness of the introduced volatility measures.

Next, we continue by assessing whether the gains of using option-implied volatility measures also extend to volatility density forecasting. Forecasting the distribution of volatility has recently received increased attention for several reasons. First of all, when forecasting returns, any uncertainty in the volatility estimates transfers to the uncertainty in the return forecasts. Knowledge about the uncertainty around the volatility point forecast is therefore valuable. Second, volatility density forecasts are gaining importance due to growing use of volatility-based estimators. For instance, Corradi et al. (2009) propose a model-free estimator for the predictive density of integrated variance. In our paper, we follow the framework of Corsi et al. (2008) to construct volatility density forecasts.

In an empirical analysis, we first assess whether the standard HAR assumption of Gaussian, independent and identically distributed innovations holds. Our results indicate that the innovations clearly deviate from normality, and that there is strong autocorrelation in the level and variance of the residuals up to at least one month. Following these empirical findings, we extend the standard HAR model with a GARCH component and skewed-t distributed residuals. Using different evaluation metrics, we show that explicitly modeling this alternative distributional assumption leads to significant gains in density forecasting. Furthermore, we show that the gains of adding option-based measures to the conventional return-based volatility estimators also extend to density forecasting. However, the forecasting gains only become clear when explicitly modeling the non-normality and time-varying variance of the HAR innovations or when specifying a less noisy volatility measure as forecasting target.

Related Literature

The volatility of asset returns is persistent, and effective analysis requires capturing long-memory dynamics (Bollerslev and Mikkelsen, 1996). Initial volatility modeling approaches acknowledging this stylized fact include the Integrated GARCH by Bollerslev (1986), the Component GARCH by Lee and Engle (1993) and the Fractionally Integrated GARCH by Baillie (1996). Upon increasing availability of intraday return data, Andersen and Bollerslev (1998) introduced the realized volatility (RV). This nonparametric volatility measure uses squared intraday returns to consistently estimate volatility, and is substantially less noisy than conventional measures based on squared daily returns. The introduction of nonparametric volatility measures uniquely enabled volatility to be observed as a time series, which in turn allowed direct forecasting of volatility through econometric models.

An alternative approach to measuring volatility is through prices of options written on the underlying asset. The foundation of this concept originates from the work of Black and Scholes (1973), who directly map option prices to the unobserved volatility of an asset, resulting in a volatility estimator termed Black–Scholes implied volatility ($BSIV$). The Black-Scholes model, used to derive this estimator, assumes constant volatility and ignores the presence of price jumps. As these assumptions have been shown to be violated in practice, the practical value of the estimator remains limited. A more popular and realistic alternative in volatility forecasting is the 'model-free' VIX index, derived from S&P 500 index options. The index reflects the market's expectation of the 30-day ahead risk-neutral variance. However, as discussed by Jiang and Tian (2007), the VIX index suffers from approximation errors that are partially predictable. More recently, the increasing liquidity and availability of short-maturity options have allowed the

construction of nonparametric volatility measures. Todorov (2019) derives such a nonparametric estimator of spot volatility using short-dates options with a variety of strike prices. Notably, this estimator provides a measure of true underlying spot volatility, and is substantially less biased than the *BSIV* or *VIX* index.

In the domain of forecasting nonparametric volatility measures, the HAR model introduced by Corsi (2009) is considered the dominant modeling approach. It is parsimonious, easy to estimate and effectively captures the long memory of volatility estimators. As a result of its strong forecasting performance, HAR models have been used and extended by a variety of studies (see, e.g., Andersen et al. (2007); Bollerslev et al. (2006); Bollerslev et al. (2016); Patton and Sheppard (2015)). However, all of these studies define linear HAR structures which assume Gaussian, independent and identically distributed innovations. Corsi et al. (2008) show that these assumptions, even in common logarithmic volatility specifications, empirically fail. Allowing for non-Gaussianity and time-varying variance of volatility can therefore improve forecasting performance, and lead to better representation of uncertainty around point forecasts, which is of key importance in constructing density forecasts.

The remainder of this paper is structured as follows. Section 2 presents the data that is used to conduct empirical analysis. Section 3 outlines the construction of option-based and return-based volatility measures, and presents an empirical evaluation of their predictive performance in point forecasting. Next, we investigate the role of different model specifications using shrinkage techniques, and assess whether the gains in volatility measurement translate into gains for density forecasting. Finally, Section 4 concludes the paper with a summary of the main findings.

2 Data

Todorov and Zhang (2022) have made their data, consisting of the high-frequency return and option-based volatility measures of the S&P 500 index, publicly accessible via the Journal of Applied Econometrics Data Archive. This paper utilizes the provided data to replicate results and extend the original analysis. Additionally, we collect *VIX* index data from the Chicago Board Options Exchange (CBOE). Daily market return data is obtained from the Kenneth R. French Data Library. The entire collected sample runs from 3 January 2008 until 31 December 2018 and covers $n = 2758$ trading days.

3 Volatility Measurement and Forecasting with Option Data

We introduce a general framework for the price process of a single asset. For some $t \geq 0$, we assume that the log-price process x_t follows the stochastic equation:

$$x_t = x_0 + \int_0^t \mu_s ds + \int_0^t \sigma_s dW_s + \sum_{s \leq t} \triangle x_s, \tag{1}$$

where μ_t denotes the drift term of the price movement, σ_s denotes the instantaneous volatility at time s, and W_t denotes a standard Brownian motion. We essentially define the price process of an asset through three separate parts: (i) a deterministic direction of the price movement over time, (ii) a stochastic term capturing random market fluctuations over time, and (iii) a term that separates discrete jumps from the continuous price process.

3.1 Volatility Measures

Return-Based Measures

We start by introducing volatility measures based on high-frequency return data. During a business day, a unit interval $[t-1, t]$ can be split up into an overnight period $[t-1, t-1+k]$ and a trading period $[t-1+k, t]$. During the trading period, we have $n+1$ equidistant price observations, and thus n intraday returns. A common aim in the volatility forecasting literature is to forecast the integrated variance (IV). The integrated variance normalized over a trading period can be defined by $\int_{t-1+k}^{t} \sigma_s + \sum_{s\in[t-1+k,t]} \triangle x_s$. Although IV is not observable, Andersen and Bollerslev (1998) propose the corresponding realized volatility (RV) estimator on a business day t as:

$$RV_t = \sum_{i=1}^{n} r_{t,i}^2, \quad t \in N_+, \tag{2}$$

where the intraday returns are defined by $r_{t,i} = x_{t-1+k+(1-k)i/n} - x_{t-1+k+(1-k)(i-1)n}$ for $t \in N_+$ and $i = 1, \ldots, n$. RV is a consistent estimator of the integrated variance as the number of intraday observations increases, i.e. when $n \to \infty$. However, the number of observations are in practice limited due to data availability.

A key limitation to the realized volatility estimator is that it does not separate continuous variation with discontinuous jumps, which are often present in the price process of an asset. Therefore, sudden large movements in the price can lead to biased estimates of volatility in finite samples. To this extent, Mancini (2009) proposes an estimator that separates continuous-path variation from discontinuous jumps:

$$TV_t = \sum_{i=1}^{n} (r_{t,i}^2) 1(|r_{t,i}| \leq 3\sqrt{min(BV_t, RV_t)} \times n^{-1/2}), \quad t \in N_+, \tag{3}$$

where $BV_t = \frac{\pi}{2}\frac{n}{n-1}\sum_{i=2}^{n} |r_{t,i}||r_{t,i-1}|$ denotes the bi-power variation measure of Barndorff-Nielsen and Shephard (2004). Intuitively, if we assume that $r_{t,i} \overset{\text{iid}}{\sim} N(0, \sigma^2/n)$, then it follows that $E[|r_{t,i}||r_{t,i-1}|] = (\sqrt{\frac{2}{\pi}\frac{\sigma^2}{n}})^2 = \frac{2}{\pi}\frac{\sigma^2}{n}$. The given finite-sample correction term $\frac{\pi}{2}\frac{n}{n-1}$ therefore results in the bi-power variation measure being an unbiased estimator of the underlying spot volatility. BV_t is used as a truncation level to filter out jumps in Equation (3), as it typically attains lower values then RV_t in the presence of jumps.

Option-Based Measures

We continue by introducing volatility measures based on OTM options written on the underlying asset. We denote $O_{t,\tau}(k)$ as the theoretical price of an OTM option observed at time t, expiring at time $t+\tau$ with log-strike price k. In practice, these option prices are observed with error, so instead $\hat{O}_{t,\tau}(k)$ is available. To ensure that the option pricing is free of arbitrage, we have that $k \leq log(F_{t,\tau})$ for a put and $k \geq log(F_{t,\tau})$ for a call, where $F_{t,\tau}$ denotes the futures price of the underlying asset at time t, which expires at time $t+\tau$. In other words, we eliminate arbitrage by pricing options relative to the discounted risk-neutral expectation of the futures price.

Given an asset's log-return $x_{t+\tau} - x_t$, we want to infer spot volatility using available option prices written on the underlying asset. We first note that, to encode the probability distribution

of some random variable X, we can estimate its corresponding characteristic function $\phi_X(u) = E[e^{iuX}]$. Consequently, all moments and density functions of X can be derived from such a characteristic function. In our case, defining $X = x_{t+\tau} - x_t$ yields us the characteristic function of the price increment, which is defined as $E_t^Q[e^{iu(x_{t+\tau}-x_t)}]$ under risk-neutral, arbitrage-free valuation Q of future asset prices. As noted by Todorov (2019), this characteristic function can be nonparametrically estimated using a set of OTM options based on a Fourier transformation:

$$\mathcal{L}_{t,\tau}(u) = 1 - (u^2 + iu) \int_R e^{(iu-1)k - iux_t} O_{t,\tau}(k) dk, \quad u \in R. \tag{4}$$

In practice, $\hat{\mathcal{L}}_{t,\tau}(u)$ is estimated with a Riemann-sum approximation based on the available strike prices k in the option portfolio. In summary, the above procedure thus allows us to infer the characteristic function of price increments through option prices. Rather than imposing a specific model structure for the price increment of an asset, we define the characteristic function of log-returns and estimate it nonparametrically using the available observed option prices.

Given an estimate of the characteristic function $\hat{\mathcal{L}}_{t,\tau}(u)$, Todorov (2019) proposes the following volatility estimator derived from the estimated characteristic function $\hat{\mathcal{L}}_{t,\tau}(u)$:

$$\hat{V}_{t,\tau}(\hat{u}_{t,\tau}) = -\frac{2}{\tau \hat{u}_{t,\tau}^2} log|\hat{\mathcal{L}}_{t,\tau}(\hat{u}_{t,\tau})|, \tag{5}$$

where $\hat{u}_{t,\tau} = min(\hat{u}_{t,\tau}^{(1)}, \hat{u}_{t,\tau}^{(2)})$. Similar to defining a truncation level in Equation (3), the choice of u ensures robust estimation during sudden price jumps. Intuitively, the choice of $\hat{u}_{t,\tau}$ defines a trade-off between reliable inference of $\hat{\mathcal{L}}_{t,\tau}(u)$ and robustness to jumps in $|\hat{\mathcal{L}}_{t,\tau}(u)|$ during estimation. The exact calculation of $\hat{u}_{t,\tau}^{(1)}$ and $\hat{u}_{t,\tau}^{(2)}$, where this trade-off is analytically specified, can be found in Todorov (2019). In this paper, it is also shown that $\hat{V}_{t,\tau}(\hat{u}_{t,\tau})$ is a consistent estimator of the true real-time underlying spot volatility. This is in contrast to return-based estimates, which rely on returns over a time window and thus estimate integrated volatility rather than instantaneous spot volatility.

On each day, we have a variety of options available that each have different maturities. The prices of the options with the shortest maturities reflect most directly the expectation of the instantaneous volatility of the underlying asset at that time. Therefore, the first option-based estimator takes a simple average of the two options with lowest maturity available:

$$OV_t = \frac{1}{2}(\hat{V}_{t,\tau_1}(\hat{u}_{t,\tau_1}) + \hat{V}_{t,\tau_2}(\hat{u}_{t,\tau_2})). \tag{6}$$

The option-based and return-based volatility estimators are asymptotically uncorrelated as their measurement errors stem from distinct sources (Todorov and Zhang, 2022). We may thus improve the measurement of volatility by combining the return-based and option-based estimators. Todorov (2019) shows that we can diversify the measurement error between the two volatility measures by combining them. More precisely, they propose the combination:

$$EV_t = \omega_t^{(1)} \hat{V}_{t,\tau_1}(\hat{u}_{t,\tau_1}) + \omega_t^{(2)} \hat{V}_{t,\tau_2}(\hat{u}_{t,\tau_2}) + \omega_t^{(3)} TV_t, \tag{7}$$

where the optimal weights are based on asymptotic variances and are formally defined in Todorov and Zhang (2022). Intuitively, the weighting scheme assigns a higher weight to volatility

measures with a relatively low asymptotic variance. The scheme allows for time-varying weights, therefore acknowledging estimator's time-varying precision in recovering spot volatility.

Efficient Realized Volatility

We conclude by introducing a less noisy proxy for future volatility, improving upon the conventional realized volatility (RV) estimator. The volatility estimator EV measures real-time diffusive spot volatility, but does not separate continuous price changes from discontinuous jump variation. By adding the difference term RV-TV, we isolate sudden discontinuous shocks from the diffusive price process to get a cleaner proxy for total quadratic variation on a day:

$$ORV_t = EV_t + RV_t - TV_t. \tag{8}$$

This efficient measure thus combines the optimal diffusive estimate EV with a return-based measure of discontinuous price jumps. Overall, it yields us a cleaner proxy for future volatility and is particularly relevant as a forecasting target in our empirical analysis.

Summary Statistics

We present summary statistics of the introduced return-based and option-based volatility measures in Appendix A. The return-based estimator RV exhibits the highest overall standard deviation, suggesting that it is substantially more noisy than all other estimates. A key limitation of RV is that it does not separate discontinuous jumps from a continuous price process. The truncated variation TV, which filters out extreme returns, displays a lower time series mean and standard deviation. We further note that the option-based measure OV has a substantially lower time series mean than that of the return-based measures. Also, the standard deviation of OV is about half the magnitude of that of TV, suggesting a less noisy and more persistent volatility estimate. The high first-order autocorrelation coefficient of OV confirms this finding. The differences between the option-based estimator OV and the optimally combined measure EV are relatively small. However, the slight reduction in standard deviation suggests a succesful reduction in noise through diversification of the measurement error. Finally, we note that the VIX index exhibits a considerably high time series mean compared to all other measures. This confirms the findings of prior literature, which suggest that the VIX index contains an upward bias, mainly due to the jump risk premia contained in option prices.

3.2 Volatility Forecasting

We continue by incorporating the introduced measures in a volatility forecasting framework proposed by Corsi (2009). We define the h-day-ahead realized volatility as $RV_{t,t+h}$ and some measure of volatility at time t as V_t. A general HAR model for a single predictor V, which we refer to as HAR-V, is then given by:

$$RV_{t,t+h} = \beta_{0,h} + \beta_{d,h}V_t + \beta_{w,h}V_{t-5,t} + \beta_{m,h}V_{t-22,t} + \epsilon_{t+h}, \tag{9}$$

where $RV_{t,t+h} = \sum_{j=1}^{h} RV_{t+j}$ and $V_{t_1,t_2} = \sum_{t=t_1+1}^{t_2} V_t$ for any volatility measure V_t. The HAR model is straightforward to estimate, and it adequately captures the long-memory property of volatility by specifying volatility components over common trading horizons.

In line with Todorov and Zhang (2022), we consider for V the return-based measures RV and TV, the option-based measure OV and the combined measure EV in Equation (9). Furthermore, we propose a HAR-MV model that directly combines the option-based and return-based measures OV and TV in the regression equation:

$$RV_{t,t+h} = \beta_{0,h} + \beta_{d,h}OV_t + \beta_{w,h}OV_{t-5,t} + \beta_{m,h}OV_{t-22,t} + \gamma_{d,h}TV_t + \gamma_{w,h}TV_{t-5,t} + \gamma_{m,h}TV_{t-22,t} + \epsilon_{t+h}. \tag{10}$$

The key difference of the HAR-MV model, compared to the HAR-EV specification, is that we do not restrict the weights of the option-based and return-based volatility measures. At the same time, this specification does require the estimation of more regression parameters. In summary, both specifications allow us to combine the option-based and return-based measures, resulting in a diversification of the measurement error. Consequently, they allow us to assess whether measurement gains translate into forecasting gains.

We evaluate the out-of-sample forecasting performance through both a Rolling Window (RW) and Increasing Window (IW) approach. In this paper, the length of the rolling window is set to 1000 days. The out-of-sample forecasting period runs from December 2011 to December 2018, covering a total of 1758 trading days. Using the classic OLS loss function for the estimation of the specified models is sub-optimal under the presence of outliers in the RV series. Therefore, one may consider a more robust loss function when estimating the models. Specifically, we employ the Huber loss function (Huber, 1992) for robust inference. The loss function is quadratic in small losses and linear in larger losses. More precisely:

$$\mathcal{L}(\phi) = \begin{cases} \mathcal{L}^2/2, & \text{if} \quad |\mathcal{L}| \leq \phi, \\ \phi|\mathcal{L}| - \phi^2/2, & \text{if} \quad |\mathcal{L}| > \phi, \end{cases} \tag{11}$$

for some tuning parameter $\phi > 0$ defining the trade-off between bias and outlier robustness.

To assess and compare the out-of-sample forecasting performance of the individual models, forecast combinations and shrinkage techniques, we employ two popular loss functions:

$$MSE = \frac{1}{q}\sum_{t=m+1}^{m+q}\left(RV_t - \hat{RV}_t\right)^2, \tag{12}$$

$$QLIKE = \frac{1}{q}\sum_{t=m+1}^{m+q}\left(\frac{RV_t}{\hat{RV}_t} - log\frac{RV_t}{\hat{RV}_t} - 1\right), \tag{13}$$

where RV_t denotes the actual realized volatility value at day t, $\hat{RV}_t$ its corresponding candidate forecast, m the size of the in-sample estimation period and q the size of the out-of-sample evaluation period.

Next, we turn to evaluating the out-of-sample forecasting losses of the different models. In Appendix B, we denote the ratios of forecasting losses for each model relative to the losses of the HAR-RV model. First, we note that the HAR-RV model performs very similar to the HAR-TV model in terms of forecasting losses. In stochastic volatility models where the intensity of

price jumps depends solely on the diffusive part of volatility, and not on a discontinuous jump process, it should be beneficial to use only a smooth volatility measure for forecasting. However, if discontinuous jumps contain predictive ability for future price jumps, it can be beneficial to include past jump variation as a predictor. Overall, HAR-RV and HAR-TV provide very similar results, suggesting that only using the smooth diffusive volatility process (TV) for forecasting is not necessarily advantageous.

The OV estimator provides a more precise measure of diffusive spot volatility compared to the return-based estimators, and we would therefore expect these gains to also extend to forecasting. Indeed, the HAR-OV model almost consistently outperforms the HAR-TV model, except for the QLIKE mean statistic of the increasing window.

Since the return-based measure TV and option-based measure OV are asymptotically uncorrelated, further precision gains in measuring spot volatility should be achieved by combining the measures. Based on the results in the table, these gains clearly transfer to forecasting. The HAR-MV model strongly improves upon the HAR-TV model, exhibiting gains up to 30% in the median forecasting losses. The HAR-EV model, which directly combines the return-based and option-based measures based on asymptotic variance, shows similar and occasionally slightly worse results compared to the HAR-MV specification. OV and TV each have a time-varying bias in measuring spot volatility, so treating them as separate predictors instead of through a weighted average can be beneficial for forecasting. In addition, the weights in EV might suffer from noise in the asymptotic variance estimation, which consequently degrades forecasting performance.

The conclusions we have made extend to all the forecasting horizons reported in Panel A, B and C. In summary, diversification of the measurement error between option-based and return-based volatility estimators substantially improves precision in measuring spot volatility, and these gains clearly extend to forecasting future volatility.

Table 1: ORV forecasting results for the S&P 500 index

Window	**Criterion**	**1 day ahead**		**1 week ahead**		**1 month ahead**	
		HAR-MV	**HAR-EV**	**HAR-MV**	**HAR-EV**	**HAR-MV**	**HAR-EV**
Rolling window	MSE (mean)	0.7473	0.7233	0.7416	0.7248	0.8815	0.8822
	QLIKE (mean)	0.5991	0.5732	0.7216	0.7280	0.8899	0.9418
Increasing window	MSE (mean)	0.6790	0.6668	0.6857	0.6747	0.7713	0.7356
	QLIKE (mean)	0.5140	0.5062	0.6378	0.6310	0.8219	0.8146

Note: The table entries are the ratios of out-of-sample forecasting losses for different models relative to the HAR-TV forecasting losses when forecasting future ORV.

We extend our analysis by comparing the best-performing HAR-MV and HAR-EV models to the benchmark HAR-TV model, when forecasting a more accurate proxy of volatility. We replace $RV_{t,t+h}$ with $ORV_{t,t+h}$ as defined in Equation (8) for the above specified forecasting models. The forecasting gains of combining option-based and return-based volatility proxies should become even clearer when forecasting a less noisy volatility proxy, as these gains might partially be hidden when forecasting noisier volatility proxies. We present the forecasting results in Table 1. Indeed, forecasting gains become even more profound when forecasting ORV. For both the HAR-MV and HAR-EV models, across each forecasting horizon, the ratios of forecasting losses

drop substantially further compared to forecasting RV.

We continue our analysis by introducing a measure of jump variation derived from option data. Until now, we have only used the information content of options to measure diffusive spot volatility. Todorov (2019) propose, however, an estimator that extracts a risk-neutral jump variation component from option data. Referring to Equation (1), this measure thus extracts the remaining information present in option data. Specifically, the option-based jump-variation component is isolated by subtracting the measure of real-time spot volatility of Equation (5) from the total risk-neutral return variation. This difference is nonparametrically estimated using the two options with lowest maturity. A simple average of these two estimates, similar to Equation (6), yields the jump variation measure JV. We leave the exact calculation to Todorov and Zhang (2022).

To assess potential forecasting benefits of this jump component, the HAR-MV model is augmented with the option-based jump component. Measurement errors in volatility estimators average out over a longer horizon, causing the weekly and monthly measurement errors to be substantially smaller than the daily errors. Therefore, the largest diversification gains in combining measurement errors are obtained through combining the daily estimates. To this extent, only the past day value of JV is included in the HAR-MV-JV specification:

$$\begin{aligned} RV_{t,t+h} = \beta_{0,h} + \beta_{d,h}OV_t + \beta_{w,h}OV_{t-5,t} + \beta_{m,h}OV_{t-22,t} + \gamma_{d,h}TV_t + \gamma_{w,h}TV_{t-5,t} + \gamma_{m,h}TV_{t-22,t} \\ + \beta_{j,h}JV_t + \epsilon_{t+h}. \end{aligned} \tag{14}$$

Todorov and Zhang (2022) show that the option-derived jump component is not significantly related to future volatility. However, potential forecasting gains may still arise through diversification of the measurement error between OV and JV. In Table 2, we compare the forecasting performance of the HAR-MV and HAR-MV-JV models in forecasting future RV. According to both evaluation criteria, the forecasting performance slightly improves across all horizons by adding the measure of jump variation. Still, we conclude that the most relevant information contained in the short-dated option portfolio is the part that recovers diffusive spot volatility. To finalize this section, we provide a brief illustration of the gains of adding option data in predicting market returns. This analysis is reported in Appendix C.

Table 2: Forecasting results using multiple option predictors for the S&P 500 index

Window	**Criterion**	**1 day ahead**		**1 week ahead**		**1 month ahead**	
		HAR-MV	**HAR-MV-JV**	**HAR-MV**	**HAR-MV-JV**	**HAR-MV**	**HAR-MV-JV**
Rolling window	MSE	0.8679	0.8310	0.8783	0.8042	0.9659	0.9453
	QLIKE	0.8112	0.7582	0.7991	0.7409	0.9233	0.9149
Increasing window	MSE	0.8805	0.7740	0.8683	0.8184	0.9755	0.9598
	QLIKE	0.7710	0.7024	0.8618	0.8250	0.9811	0.9795

Note: The table entries are the mean relative losses of HAR-MV and HAR-MV-JV models with respect to that of the HAR-TV model.

3.3 Shrinkage Methods

Combining option-based and return-based volatility measures leads to more precise measurement of volatility, resulting in improved forecasting ability. It should be noted, however, that the measurement errors of all volatility proxies in recovering true spot volatility are time-varying and arise from different sources. Therefore, it is not clear whether a single volatility measure consistently provides dominant predictive ability for future volatility at each point in time. Including daily, weekly, and monthly lags of all option-based proxies (OV, JV, EV, VIX) alongside return-based measures (RV, TV) may therefore yield additional forecasting gains, especially if different predictors (or combinations of predictors) matter at different times. It is, however, not clear whether a model with a large amount of predictors would generalize well out-of-sample.

To assess this trade-off, we evaluate shrinkage techniques that adaptively select and weigh predictors. Shrinkage methods aim at selecting the most useful predictors from a large set of regressors by penalizing regression coefficients. If forecasting gains are extractable from shrinkage methods over simpler specifications, this indicates that adaptive selection from the full set of regressors, based on in-sample performance, can improve forecasting performance. In other words, it would suggest that individual measures do not provide equal contribution to forecasting, and selecting a subset of measures at each point in time is favorable. In contrast, no forecasting gains over simpler specifications would suggest that including the additional volatility measures is of limited relevance. This indicates that although the measurement error in recovering spot volatility is time-varying for all estimators, it does not translate into meaningful variation in the predictive ability of individual regressors.

In this section, we focus on forecasting the less noisy volatility proxy ORV, as this should make any forecasting gains more clear. We compare the results of the shrinkage methods to the forecasting results of Table 1. Following Zhang et al. (2019), we create a general predictive model for the h-day ahead ORV that includes the daily, weekly and monthly lags of the aforementioned volatility measures as predictors:

$$ORV_{t,t+h} = \beta_{0,h} + \sum_{i=1}^{q} \beta_{i,h} f_{i,t} + \epsilon_{t+h}, \tag{15}$$

where $f_{i,t}$ denotes the i-th predictor available at the end of day t, q represents the total number of predictors, and ϵ_{t+h} denotes an error term.

A common way to implement shrinkage is through penalized regression, which shrinks some coefficients towards zero, thereby simultaneously introducing some bias and decreasing the variance of the estimates (Tibshirani, 1996). More precisely, a general penalized estimation for Equation (15) is given by:

$$\beta_h = \arg\min_{\beta_h} \left(\underbrace{\sum_{t=1}^{T} \left(ORV_{t,t+h} - \beta_{0,h} - \sum_{i=1}^{q} \beta_{i,h} f_{i,t} \right)^2}_{\text{OLS Loss Function}} + \underbrace{\lambda \sum_{i=1}^{q} |\beta_{i,h}|^p}_{\text{Penalty Function}} \right). \tag{16}$$

Within this estimation framework, we consider three widely used penalty functions. Setting $p = 1$ yields the lasso estimation by Tibshirani (1996), which shrinks some coefficients towards

zero so that the most relevant predictors are selected. Besides, setting $p = 2$ results in the ridge penalty of Hoerl and Kennard (1970), which shrinks coefficients but retains all predictors. Finally, a convex combination of the lasso and ridge penalties given by $\lambda \sum_{i=1}^{q}(\alpha|\beta_{i,h}| + (1-\alpha)\beta_{i,h}^2)$ yields the elastic net penalty of Zou and Hastie (2005), which performs particularly well when predictors are highly correlated. Note that Equation (16) is expressed using the standard OLS loss function. However, regular shrinkage methods may not perform well in the presence of outliers. To ensure robust estimation, we instead consider the Huber loss function as defined in Equation (11). We thus combine three well-known penalty functions (ridge, lasso, elastic-net) with the Huber loss function to estimate Equation (15).[1]

An important choice in all penalized regressions is the tuning parameter λ, which defines the trade-off between in-sample fit and model complexity. We follow a time-series cross-validation scheme, in which we split up the training data of the current window iteration into K consecutive, non-overlapping subsets. Then, for each subset $k \in \{2, \ldots, K\}$, we estimate the model on the previous $k-1$ subsets and evaluate its forecasting performance on the k-th subset. We compute the average prediction error of this process across the $K-1$ evaluated subsets. This entire scheme is repeated over a predefined grid of lambda values, and the lambda value yielding the lowest average prediction error is selected.[2]

Table 3 shows the out-of-sample forecasting results of the ridge, lasso and elastic net methods, and compares them to the previous HAR-MV and HAR-EV specifications from Table 1. The results reveal that the shrinkage methods do not manage to consistently outperform the simpler HAR-MV and HAR-EV specifications across the forecasting horizons and evaluation metrics. For the increasing window, the HAR-EV specification still yields the lowest QLIKE mean statistic at the 1-month forecasting horizon. Similarly for the rolling window, there are no gains in the MSE mean statistic over the HAR-EV specification at the 1-day forecasting horizon. The results suggest that although the measurement errors of volatility proxies stem from different sources and vary over time, additional gains of selecting the most relevant combination of measures at each point in time are limited. Furthermore, the additional regressors do not appear to offer much unique predictive information, as most of their signal appears to be already captured by the combination of OV and TV. Also, it is not directly obvious whether the forward-looking VIX index captures additional information about future volatility not already reflected in the constructed measures. We conclude that the additional forecasting benefits of using the VIX index are limited, highlighting that the constructed measures prove as effective predictors. In summary, combining the option-based measure OV and return-based measure TV is surprisingly effective in a volatility forecasting model. For forecasting purposes, our results point towards

[1] We employ three basic shrinkage techniques. For empirical forecasting applications using more advanced methods such as the adaptive LASSO, group LASSO and ordered LASSO, we refer to Audrino et al. (2015), Audrino and Knaus (2016) and Ding et al. (2021). Our choice is motivated by three considerations: (i) this study focuses on the trade-off between parsimonious and adaptive model specifications, rather than identifying the optimal shrinkage method, (ii) combining advanced shrinkage techniques with a Huber loss function complicates numerical estimation substantially and is not directly supported by standard packages, and (iii) as noted by Zhang et al. (2019), using many advanced shrinkage methods raises concerns about data mining.

[2] In this paper, we set $K = 5$. Note that for each fold $k \in \{2, \ldots, K\}$, the Huber loss function is applied to ensure robust in-sample estimation of the preceding $k-1$ folds in the presence of outliers. However, the validation on the k-th fold, which is used to select the tuning parameter λ, is still based on the mean squared error (MSE). This choice aligns the validation process with the loss function (MSE) used to evaluate the actual out-of-sample forecasts. Finally, the lambda grid is obtained as the default sequence created by the `hqreg` package in R.

the usage of a parsimonious HAR specification with a combination of the option-based OV and return-based TV as predictors.

Table 3: ORV forecasting results for the S&P 500 index using shrinkage methods

Window	Criterion	HAR-MV	HAR-EV	ENet	Ridge	LASSO
Panel A: 1-day ahead relative forecasting loss						
Rolling window	MSE (mean)	0.7473	0.7233	0.7237	0.7305	0.7237
	QLIKE (mean)	0.5991	0.5732	0.5632	0.5651	0.5634
Increasing window	MSE (mean)	0.6790	0.6668	0.6642	0.6592	0.6633
	QLIKE (mean)	0.5140	0.5062	0.4990	0.4918	0.4995
Panel B: 1-week ahead relative forecasting loss						
Rolling window	MSE (mean)	0.7416	0.7248	0.7305	0.7228	0.7300
	QLIKE (mean)	0.7216	0.7280	0.6786	0.6799	0.6784
Increasing window	MSE (mean)	0.6857	0.6747	0.6736	0.6999	0.6734
	QLIKE (mean)	0.6378	0.6310	0.6336	0.6552	0.6297
Panel C: 1-month ahead relative forecasting loss						
Rolling window	MSE (mean)	0.8815	0.8822	0.8762	0.8725	0.8796
	QLIKE (mean)	0.8899	0.9418	0.9294	0.9384	0.9317
Increasing window	MSE (mean)	0.7713	0.7356	0.7171	0.7264	0.7339
	QLIKE (mean)	0.8219	0.8146	0.8195	0.8187	0.8383

Note: The entries in the table are the ratios of the out-of-sample forecasting losses for different models relative to the HAR-TV benchmark when forecasting future ORV. For the elastic net penalty, we follow a conventional choice in the literature by setting $\alpha = 0.5$, which limits computational cost as we do not require extra parameter tuning.

3.4 Density Forecasting

We now extend our analysis by quantifying the gains of adding option-implied volatility measures in density forecasting. Specifically, we evaluate whether the accuracy of density forecasts improves when using option-augmented HAR models over return-only HAR models. Given that the shrinkage methods in Section 3.3 did not substantially improve the accuracy of point forecasts, our focus will be on the simpler HAR-TV, HAR-MV and HAR-EV specifications that we have previously analyzed. We only focus on one-day-ahead forecasts to maintain a link to daily trading horizons and to reduce computational costs.[3] We denote a set of n one-day-ahead density forecasts for y as $\{f_{t|t-1}(y_t)\}_{t=1}^{n}$, in which the density forecast is constructed from a parametric forecasting model using information up to time $t-1$.

In practice, density forecasts provide a full characterization of future return uncertainty. In other words, density forecasts model the uncertainty around volatility itself. In risk management applications, uncertainty around volatility forecasts transmits to uncertainty in return forecasts. Metrics such as Value-at-Risk (VaR) and Expected Shortfall (ES), implied by distribution forecasts, may therefore be particularly relevant. Furthermore, correctly specified volatility density forecasts can also be important in derivative pricing, especially as volatility-based derivatives become more widespread and require accurate pricing. Given the practical value of density forecasts, it is relevant to integrate option-implied volatility measures to improve density forecasts.

[3]The existing literature places a strong emphasis on producing density forecasts at the daily horizon, which is partially due to the practical implications of daily forecasts (see, e.g., Corradi et al. (2009); Corsi et al. (2008)). The introduced methodology in this paper can, however, also be extended to multi-horizon density forecasts, but this requires substantial computing power. In this paper, the multi-horizon analysis is omitted due to computational and time constraints.

Empirical Analysis

To provide a full predictive distribution around a point forecast, it is essential to correctly specify the distribution of the residuals resulting from a forecasting model. Incorrect residual assumptions lead to a misrepresentation of the actual uncertainty around the forecast. In our study, the HAR-type models are linear structures, which assume Gaussian, independent and identically distributed (IID) error terms. However, empirical findings such as that of Corsi et al. (2008) and Andersen et al. (2001) show that these assumptions are frequently violated in practice. As possible residual misspecifications can heavily degrade the quality of density forecasts, we first conduct an in-sample analysis of the HAR residuals resulting from the forecasting models of interest. Since all models in this paper are estimated using robust Huber estimation, we report some in-sample diagnostic statistics of the Huber residuals in Table 4.

Table 4: Residual Diagnostics for Huber-Estimated HAR Models

Model	**Skewness**	**Kurtosis**	**JB**
HAR-TV	13.192	301.282	1,044,259.0***
HAR-MV	14.515	331.119	1,261,355.9***
HAR-EV	8.963	103.382	1,255,100.0***

Model	**ARCH(5)**	**ARCH(22)**	**Q(5)**	**Q(22)**	**Q2(5)**	**Q2(22)**
HAR-TV	538.950***	631.990***	522.863***	909.193***	690.056***	896.799***
HAR-MV	530.648***	661.790***	573.648***	1,147.237***	652.363***	888.845***
HAR-EV	2007.832***	2107.505***	6889.657***	13,751.853***	5444.711***	8293.613***

Note: The table reports diagnostic statistics of the residuals of the HAR-TV, HAR-MV and HAR-EV models in forecasting one-day-ahead *RV*, estimated using Huber regression. The regressions are conducted over the entire sample from 3 January 2008 until 31 December 2018. JB depicts the test statistic of the Jarque-Bera test. ARCH($\cdot$) represents ARCH-LM test statistics. Q2($\cdot$) represents the Q-statistics of the Ljung-Box test for the squared residuals. *** denotes rejection of the null hypothesis at the 1% significance level.

Table 4 shows that the estimated residuals of volatility are right-skewed and fat-tailed. For all HAR specifications, the null of normally distributed error terms is convincingly rejected by the Jarque-Bera test. The Gaussian assumption is thus likely sub-optimal in density forecasting. Besides, the ARCH-LM test statistic indicates that the conditional variance of volatility depends on past squared residuals. In other words, the test indicates strong autoregressive conditional heteroskedasticity. The Q-statistics of the Ljung-Box test indicate substantial autocorrelation in both the level and variance of the residuals up to at least one month. Combined with the ARCH-LM test statistics, the results reveal significant serial dependence in the first and second moment of the HAR-residuals, indicating a violation of the IID assumption.

A common modeling alternative, which will also be adopted in this paper, is to apply a logarithmic transformation to the volatility forecasting target (see, e.g., Andersen et al. (2001); Corsi et al. (2008); Gonçalves and Meddahi (2009)). In practice, forecasting the logarithm of realized volatility in a HAR-type model can be expected to make the Gaussian error assumption more suitable. We report the in-sample residual diagnostics for forecasting logarithmic *RV* in Appendix D. The results show a substantial reduction in the skewness and kurtosis of the residuals, but normality is still significantly rejected by the Jarque-Bera test. The Ljung-Box and ARCH-LM tests again imply strong serial dependence in the variance of the HAR-residuals.

Our results suggest that relaxing the residual HAR-assumptions could improve the quality of density forecasts in both a standard and logarithmic volatility specification.

Alternative Distributional Assumptions

To address our empirical findings on the violation of the standard residual assumptions, we propose extensions that allow for alternative distributional assumptions in HAR specifications. Corsi et al. (2008) propose several extensions that explicitly account for the observed residual properties, and assess the gains of these modeling alternatives in density forecasting. Following their work, we start by modeling the time-varying conditional variance of the residuals by extending the HAR model with a GARCH(1,1) component:

$$\begin{aligned} y_{t,t+1} &= \beta_{0,h} + \beta_{d,h} V_t + \beta_{w,h} V_{t-5,t} + \beta_{m,h} V_{t-22,t} + \epsilon_{t+1}, \\ \sigma^2_{t+1} &= \omega + \alpha \epsilon^2_t + \lambda \sigma^2_t, \\ e_{t+1} | I_t &\sim (0,1), \end{aligned} \tag{17}$$

where the residuals are decomposed as $\epsilon_{t+1} = \sigma_{t+1} e_{t+1}$ and y denotes the forecasting target of interest (RV, ORV, log RV or log ORV). The GARCH component enables modeling time-varying variance $\{\sigma^2_{t+1}\}$ in the residuals $\{\epsilon_{t+1}\}$, allowing an improved ability to capture possible serial dependence or volatility clustering. The above specification, however, does not account for the non-normality observed in the residuals. To this extent, we specify the skewed-t distribution for $\{e_{t+1}\}$ in Equation (17) (Fernández and Steel, 1998).[4] The distribution is relatively easy to estimate, and is able to recreate a variety of asymmetric and fat-tailed distributions. Specifically, the probability density function of the skewed-t distribution is given by:

$$f_{\nu,\beta}(e_t) = \frac{2\beta}{1+\beta^2} (t_\nu(\beta e_t) I(e_t < 0) + t_v(\beta^{-1} e_t) I(e_t \geq 0)), \tag{18}$$

where t_v denotes the standard student-t density with ν degrees of freedom and $\beta > 0$ represents an asymmetry coefficient that defines the skewness of the distribution. We should note that the decomposition and modeling of the innovations in Equation (17) generalizes to HAR-specifications that include multiple volatility measures as predictors in the exact same way.

Constructing and Evaluating Density Forecasts

We assess the statistical gains of modeling time-varying residual variance and a more flexible residual distribution. Specifically, for both the HAR-TV, HAR-MV and HAR-EV specifications, we compare a benchmark HAR specification assuming Gaussian errors to a specification that assumes a GARCH component and skewed-t innovations. We retain the same point forecasts for each of the HAR models as constructed in Section 3.2, and flexibly specify the distributional structure of the forecast errors. This allows us to quantify the gains of accounting for time-varying residual variance and non-normality of the residuals in density forecasting.

[4] In addition to the skewed-t distribution, we have also experimented with specifying a student's-t distribution for the standardized residuals $\{e_{t+1}\}$ in the above framework. Appendix E reports and discusses the corresponding density forecasting accuracy measures (introduced below), showing that the student's-t distribution improves upon a Gaussian benchmark. However, the skewed-t distribution provides further accuracy gains in density forecasting beyond its symmetric counterpart.

Turning next to estimation, we use a rolling window of 1000 trading days to produce the density forecasts. The expanding window estimation is excluded to limit computational cost. For each HAR specification, the conditional density forecast for the one-day-ahead (log) RV or (log) ORV of the benchmark model is estimated as a normal distribution centered around the respective point forecast, with variance equal to the variance of the in-sample residuals of the past rolling window. Furthermore, for each HAR specification, we fit a GARCH(1,1) model with skewed-t innovations to the in-sample residuals from the past rolling window, jointly estimating both the conditional variance and the distributional parameters through maximum likelihood estimation. The density forecast is then centered around the respective point forecast, with variance equal to the one-step-ahead conditional variance forecast of the estimated GARCH model, and shaped according to the estimated skewness and tail parameters of the skewed-t distribution.[5]

In summary, we specify a standard and logarithmic volatility forecasting target, three different HAR specifications, and two different residual assumptions, overall resulting in twelve distinct sets of density forecasts. To ensure correct comparability between logarithmic and standard volatility specifications, we transform the logarithmic forecasts $X = \exp(Y)$, where $Y = \log(RV)$. This nonlinear transformation introduces a convexity bias, which we correct for by taking the log-normal density transformation $\frac{1}{y} f_{\log Y}(\log y)$, ensuring proper adjustment by the Jacobian.

A common way to evaluate the accuracy of a set of conditional density forecasts is based on the probability-integral transformation, as proposed by Diebold et al. (1998). The probability-integral transformation represents the cumulative density function of the distribution forecast evaluated at the realized value y_t:

$$z_t = \int_{-\infty}^{y_t} f_{t|t-1}(u)du. \tag{19}$$

Diebold et al. (1998) show that if the predictive density is correct, then the sequence $\{z_t\}_{t=1}^{n}$ should follow an IID Uniform(0,1) distribution.

Figure 1 presents the probability-integral transformations of the forecasting models for realized volatility (RV). It is clearly visible that the z_t of the Gaussian density forecasts are not aligned with the uniform distribution. However, the incorporation of the alternative distributional assumptions leads to density forecasts for which z_t is much closer to uniformity in all specifications. In other words, the accuracy of the density forecasts is substantially improved by allowing for more realistic assumptions of the innovations. A similar conclusion can be drawn from the probability-integral transformations of the ORV forecasts, which are presented in Appendix F.1. The results confirm, as suggested by our empirical analysis, that the Gaussian innovation assumption is sub-optimal in density forecasting. We also present the probability-integral transformations of the logarithmic volatility specification in Appendix F.2. For the Gaussian density forecasts, z_t is more similar to a uniform distribution than its non-logarithmic counterpart in Figure 1. Also, the gains from relaxing residual assumptions are less clearly

[5]Note that we essentially define a two-step estimation procedure: we first estimate volatility point forecasts using the HAR models, and subsequently estimate the conditional distribution of the resulting in-sample forecast errors. A potential extension for further research, although more complex, would be to jointly estimate the equations defined in (17).

visible in the logarithmic specification.

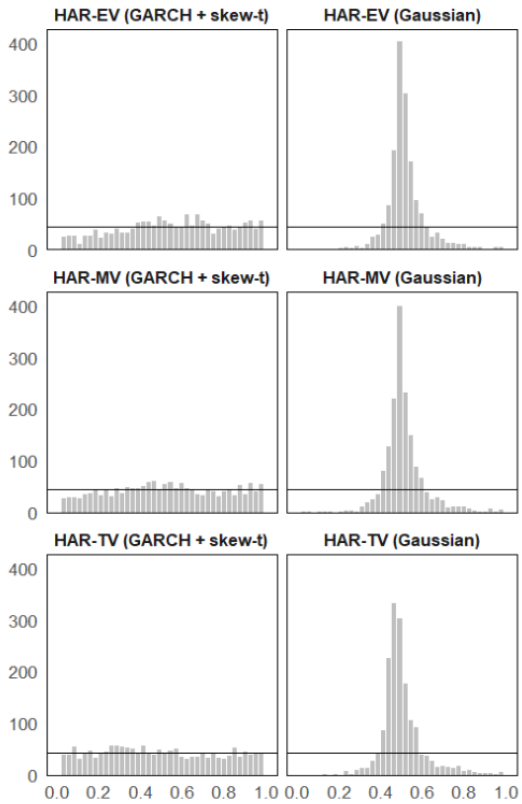

Figure 1: Probability-Integral Transformation (PIT) histograms for one-day-ahead RV density forecasts. The histograms are presented for both the HAR model with Gaussian innovations and the HAR model with a GARCH(1,1) component and skewed-t innovations.

To numerically express the visual results and statistically compare the option-augmented models over the return-only specifications, we require further analysis of the density forecasts. To this extent, we compute the logarithmic scores of density forecasts to quantify the accuracy of different forecasting models. For some one-step-ahead density forecast $f_{t|t-1}(\cdot)$ and a realized value y_t at time t, this score can be denoted as $\log(f_{t|t-1}(y_t))$. A higher score implies that the density forecast assigns larger probability mass to the observed outcome, which is favorable. Calculated over a set of density forecasts, the logarithmic scores tell us about the overall accuracy of a model's forecasts. As noted by Corsi et al. (2008), the logarithmic scores resulting from forecasting logarithmic volatility can not be directly compared to scores of the standard volatility forecasts. The scores of the logarithmic volatility models are therefore transformed as $\log(\frac{f_{t|t-1}(\log RV_t)}{RV_t})$.[6]

We apply the test of Amisano and Giacomini (2007) to assess whether the difference between the logarithmic scores of two sets of density forecasts is statistically significant. This allows us to formally compare the forecasting performance between different model specifications. For two sets of density forecasts $\{f^1_{t|t-1}(y_t)\}_{t=1}^n$ and $\{f^2_{t|t-1}(y_t)\}_{t=1}^n$ resulting from different parametric forecasting models, we compute the Log Predictive Score Difference (LPSD) as:

$$LPSD_t = \log(f^1_{t|t-1}(y_t)) - \log(f^2_{t|t-1}(y_t)), \qquad t = 1, \ldots, n. \tag{20}$$

[6]This transformation, as discussed earlier, ensures correct scaling of the scores by correcting for the change in target variable from RV to $\log RV$ (or equivalently from ORV to $\log ORV$).

The null hypothesis of an average difference of zero between the logarithmic scores of two sets of density forecasts can then be assessed using the statistic:

$$t = \frac{\overline{LPSD}}{\hat{\sigma}_n/\sqrt{n}}, \tag{21}$$

where $\overline{LPSD} = \frac{1}{n}\sum_{t=1}^{n} LPSD_t$ and $\hat{\sigma}_n^2$ denotes the heteroskedasticity and autocorrelation consistent (HAC) estimator of the asymptotic variance $\sigma_n^2 = Var(\sqrt{n} \cdot \overline{LPSD})$.[7]

We continue by quantifying the gains of explicitly modeling alternative distributional assumptions for the residuals of the HAR specifications. Table 5 reports, for each specification of interest, the comparison between density forecasts of realized volatility based on a residual GARCH component and skewed-t innovations, and its benchmark counterpart assuming Gaussian innovations. A positive t-statistic indicates that the predictive scores of the alternative specification are on average higher than those of the benchmark.

Table 5: Amisano and Giacomini (AG)-test results comparing the predictive accuracy of RV density forecasts resulting from flexible residual specifications (GARCH with skewed-t innovations) to density forecasts resulting from Gaussian residuals specifications.

Model	RV			$\log(RV)$		
	$\overline{LPSD}$	t-stat	p-value	$\overline{LPSD}$	t-stat	p-value
HAR-TV	2.3966	2.429	0.0151	0.0886	5.208	0.0000
HAR-MV	2.5806	2.085	0.0371	0.0884	4.994	0.0000
HAR-EV	2.4716	2.224	0.0261	0.0779	4.873	0.0000

Table 5 reveals that, for both the return-only and option-augmented specifications, the logarithmic scores of the alternative residual specification are significantly higher than those of the benchmark residual specification. More precisely, the t-statistics of all tests comparing the density forecasts of the alternative specification to the benchmark specification are statistically significant at the 5% confidence level. Allowing for non-normality and time-varying conditional variance clearly leads to a better representation of uncertainty around point forecasts, and thus of the entire distribution. As a result, the density forecasts assuming a more realistic underlying distribution assign higher probability mass to the observed outcome, resulting in significantly higher logarithmic scores. The statistical gains become most apparent under the logarithmic volatility specification. We note that these findings are consistent across the density forecasts for ORV and its logarithmic transformation, as reported in Appendix G.

We continue by assessing whether the gains of using option-based volatility measures also extend to density forecasting. We have shown that combining option and return data leads to improved measurement of volatility, which consequently enhances the accuracy of point forecasts for future volatility. A more precise measurement of volatility can, however, also reduce the variance and improve the stability of the in-sample residuals of a forecasting regression, which are directly used to estimate the parameters of a conditional distribution in density forecasting. More precisely, in-sample forecasting errors reflect both model misspecification and measurement noise in the volatility proxies. Filtering out measurement noise by combining option and return

[7] We follow a standard approach by computing HAC standard errors using the Newey-West estimator with automatic lag selection.

data may therefore stabilize the forecasting errors. Consequently, the forecast errors may reflect more precisely the true underlying uncertainty around the volatility forecast, thereby yielding more accurate density forecasts.

Table 6: Amisano and Giacomini (AG)-test results comparing the predictive accuracy of HAR-MV and HAR-EV density forecasts to the HAR-TV benchmark, under both Gaussian and flexible residual specifications (GARCH with skewed-t innovations).

Target	**HAR-MV**			**HAR-EV**		
	$\overline{\text{LPSD}}$	t-stat	p-value	$\overline{\text{LPSD}}$	t-stat	p-value
Gaussian innovations						
RV	-0.1614	-0.638	0.5238	-0.1086	-0.849	0.3956
log RV	0.0907	4.626	0.0000	0.1084	5.173	0.0000
ORV	-0.1348	-0.395	0.6926	-0.1259	-0.399	0.6894
log ORV	0.1920	7.361	0.0000	0.1993	8.044	0.0000
GARCH + skewed-t innovations						
RV	0.0227	1.466	0.1427	-0.0336	-1.779	0.0753
log RV	0.0905	6.613	0.0000	0.0978	6.927	0.0000
ORV	0.2206	11.850	0.0000	0.2251	11.071	0.0000
log ORV	0.1451	10.602	0.0000	0.1426	11.556	0.0000

Table 6 shows that, under the flexible residual specifications, density forecasts from the HAR-MV and HAR-EV models consistently outperform those of the HAR-TV model, except when forecasting the level of realized volatility (RV). More precisely, the AG test yields positive log predictive score differences with corresponding t-statistics that are significant at the 1% confidence level. In contrast, under the Gaussian innovations assumption, clear gains from adding option data are only apparent when forecasting logarithmic volatility measures. A potential explanation for this is that the logarithmic transformation reduces extreme variation and skewness in the forecasting target, which in turn leads to a shape of the distribution that is closer to the Gaussian assumption. Since this logarithmic transformation thus reduces misspecification in the shape of the forecasted densities, even small improvements in the respective log scores can become more clear.

As already highlighted in Figure 1, residual misspecification results in a misrepresentation of the true uncertainty around point forecasts, which potentially masks the incremental gains of adding more accurate volatility proxies. When accounting for the non-normality and time-varying variance of the residuals, the incremental gains of allowing for more accurate volatility proxies become substantially more apparent. However, the above results indicate that we still require to specify a less noisy volatility forecasting target, either through a logarithmic transformation or by adding option data (ORV), to reveal consistent incremental gains of combining option data with return data in volatility density forecasting.

4 Conclusion

This paper studies the incremental gains of adding option-based volatility measures in point forecasting and density forecasting of volatility. Following the work of Todorov (2019), we construct a nonparametric estimate of volatility using short-dated S&P 500 index options with a

wide range of strikes. As the measurement error of this option-based estimator is asymptotically uncorrelated to that of a high-frequency return-based estimator, we are able to improve measurement of volatility. In a multi-horizon forecasting framework, we empirically show that the gains in measurement transfer to gains in forecasting. Across daily, weekly and monthly forecasting horizons, combining the option-based volatility measure with a return-based measure substantially reduces the average out-of-sample forecasting loss compared to a specification solely using a return-based measure. The documented forecasting gains become even more clear when using a less noisy estimate of future volatility.

Building on this forecasting framework, we extend our analysis by evaluating whether the documented gains can be further improved when using a forecasting model that includes all introduced volatility measures. To estimate this general model, we use different shrinkage techniques that adaptively select and estimate the most relevant predictors over time. We find that although the measurement errors of different volatility proxies are time-varying, the benefit of adaptively selecting combinations of predictors at each point in time is limited. Our results point towards using a parsimonious HAR specification that combines the option-based volatility measure with a return-based measure.

Finally, we assess whether the gains of adding option-implied volatility measures to return-based measures also extend to density forecasting. We first show that relaxing the distributional assumption of the HAR innovations, by allowing for a GARCH component and skewed-t innovations, leads to a better representation of uncertainty around a point forecast. This, in turn, leads to more accurate density forecasts. When following the original HAR assumptions and coupling volatility forecasts to a Gaussian distribution, the gains of adding option data remain largely absent. We argue that the gains are likely masked due to a misrepresentation of uncertainty around the point forecasts. Only when forecasting a log-transformed volatility target do the gains become clearly visible, as the distribution of the forecast target is closer to the Gaussian shape. Under a specification with relaxed residual assumptions, we find convincing statistical evidence that the density forecasts based on a combination of option-based and return-based measures are superior to a return-only specification. However, this statistical evidence only appears when forecasting a less noisy volatility target, which is either a log-transformed volatility measure or a combined volatility measure.

We should mention some limitations of the conducted study. First of all, the shrinkage techniques rely on a limited amount of volatility proxies and assume strictly linear model structures, which likely affects our results. Incorporating more advanced measures that account for leverage and jump effects in volatility, or relaxing the linear HAR structure by allowing for nonlinearities or regime-switching behavior, should give more credit to the concept of adaptive regressor selection. Furthermore, the generalization of our density forecasting results remain limited as we only consider the daily forecasting horizon. Besides, the introduced two-step estimation procedure to produce density forecasts implicitly assumes that forecast uncertainty is fully captured in the residuals, thereby treating the estimated point forecast coefficients as fixed. A more realistic alternative would be to jointly estimate the mean and variance process.

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A Summary Statistics

Table 7: Summary Statistics of Volatility Measures

Measure	**Mean**	**STD**	**STD_D**	**Q25**	**Median**	**Q75**	**AR(1)**
RV	0.0396	0.0965	0.0724	0.0083	0.0161	0.0354	0.7188
TV	0.0281	0.0673	0.0458	0.0057	0.0113	0.0248	0.7693
OV	0.0222	0.0351	0.0121	0.0058	0.0108	0.0232	0.9405
VIX	0.0484	0.0631	0.0159	0.0181	0.0286	0.0513	0.9683
EV	0.0211	0.0331	0.0123	0.0056	0.0104	0.0219	0.9305
JV	0.0237	0.0371	0.0169	0.0078	0.0131	0.0235	0.8964

Note: The table presents summary statistics of the introduced option-based and return-based volatility measures of the S&P 500 index for the sample running from 3 January 2008 until 31 December 2018. For each measure, we report the time series mean (Mean), standard deviation (STD), standard deviation of the first difference series (STD_D), quantiles (Q25, Median, Q75), and first-order autocorrelation coefficient (AR(1)). Each volatility measure is reported in annualized variance units.

B Point Forecasting Results

Table 8: Forecasting results for the S&P 500 index

Window	Criterion	HAR-RV	HAR-TV	HAR-OV	HAR-MV	HAR-EV
Panel A: 1-day ahead relative forecasting loss						
Rolling window	MSE (median)	0.9608	1.0000	0.6698	0.6625	0.6283
	MSE (mean)	0.9891	1.0000	0.8899	0.8679	0.8677
	QLIKE (median)	0.9949	1.0000	0.7185	0.6959	0.7053
	QLIKE (mean)	0.9868	1.0000	0.9192	0.8112	0.8453
Increasing window	MSE (median)	1.0463	1.0000	0.6069	0.5823	0.5613
	MSE (mean)	0.9651	1.0000	0.9021	0.8805	0.8745
	QLIKE (median)	1.0137	1.0000	0.7139	0.6899	0.7345
	QLIKE (mean)	1.0036	1.0000	1.0379	0.7710	0.9243
Panel B: 1-week ahead relative forecasting loss						
Rolling window	MSE (median)	1.0200	1.0000	0.6326	0.6248	0.5905
	MSE (mean)	1.0270	1.0000	0.8957	0.8783	0.8817
	QLIKE (median)	0.9938	1.0000	0.7091	0.7054	0.6955
	QLIKE (mean)	1.0028	1.0000	0.9155	0.7991	0.8832
Increasing window	MSE (median)	0.9851	1.0000	0.7165	0.6559	0.6668
	MSE (mean)	0.9916	1.0000	0.8453	0.8683	0.8236
	QLIKE (median)	0.9832	1.0000	0.8873	0.7631	0.8265
	QLIKE (mean)	1.0012	1.0000	1.0099	0.8125	1.0028
Panel C: 1-month ahead relative forecasting loss						
Rolling window	MSE (median)	0.9623	1.0000	0.6836	0.8400	0.7457
	MSE (mean)	1.0126	1.0000	0.9682	0.9659	0.9723
	QLIKE (median)	0.9695	1.0000	0.8909	0.8643	0.8425
	QLIKE (mean)	1.0119	1.0000	1.0080	0.9233	1.0218
Increasing window	MSE (median)	0.9991	1.0000	0.6395	0.8737	0.7233
	MSE (mean)	0.9918	1.0000	0.8808	0.9755	0.8889
	QLIKE (median)	0.9387	1.0000	0.7402	0.8900	0.7928
	QLIKE (mean)	1.0023	1.0000	1.0173	0.9811	0.9992

Note: The table entries are the ratios of out-of-sample forecasting losses for different models relative to the HAR-TV forecast losses.

C Return Predictability Results

This section provides a brief illustration from the gains of adding option data for market return predictability. In a foundational paper, Bollerslev et al. (2009) have shown that market variance risk premium is a strong predictor for future market returns. Market variance risk premium is formally defined as:

$$VRP_t(1) = VIX_t^2 - RV_{t-22,t}, \tag{22}$$

where VIX_t denotes the monthly volatility index of the CBOE. By combining option and return data, we have created a more precise measure of spot volatility (EV), which in turn allows us to improve the measurement of realized volatility in the past month. Using this improved measure of past month realized volatility, as defined in Equation (8), we can consequently improve measurement of the variance risk premium by defining:

$$VRP_t(2) = VIX_t^2 - ORV_{t-22,t}. \tag{23}$$

We also consider a measure of jump risk premium as a predictor for future returns. To construct the jump risk premium, we subtract the daily realized jump-variation from the option-based risk-neutral jump variation. This allows us to isolate the compensation that investors demand for their exposure to jump risk in the market. The premium is averaged out over an entire month, as the daily estimate is relatively noisy. More specifically, the measure is defined as:

$$JRP_t = JV_{t-22,t} - (RV_{t-22,t} - TV_{t-22,t}). \tag{24}$$

To assess implications for return predictability, we run the following regressions:

$$r_{t,t+h} = \alpha_{0,h} + \alpha_{1,h} X_t + \epsilon_{t+h}, \tag{25}$$

where $r_{t,t+h}$ denotes market excess returns over the interval $[t, t+h]$ and X_t denotes either $VRP_t(1), VRP_t(2)$ or JRP_t. Figure 2 reveals that the original measure of variance risk premium, $VRP(1)$, yields statistically significant predictive power up to around 8 months. However, the more efficient measure $VRP(2)$ exhibits different behavior. Beneath the 6-month horizon, the respective t-statistics are below the one-sided 5% confidence level. As $VRP(2)$ serves as a significant predictor at the longer forecasting horizons, we can conclude that measurement precision of the variance risk premium alters predictive ability for excess returns. In particular, the short-term predictive ability of $VRP(1)$ may appear stronger as future measurement errors in $RV_{t-22,t}$ are partially predictable in a regression model at shorter horizons (Bollerslev et al., 2009). Finally, we note that JRP provides the highest R^2 statistics from the fifth forecasting horizon onwards, thereby outperforming risk premium statistic $VRP(2)$ at the respective long-term horizons.

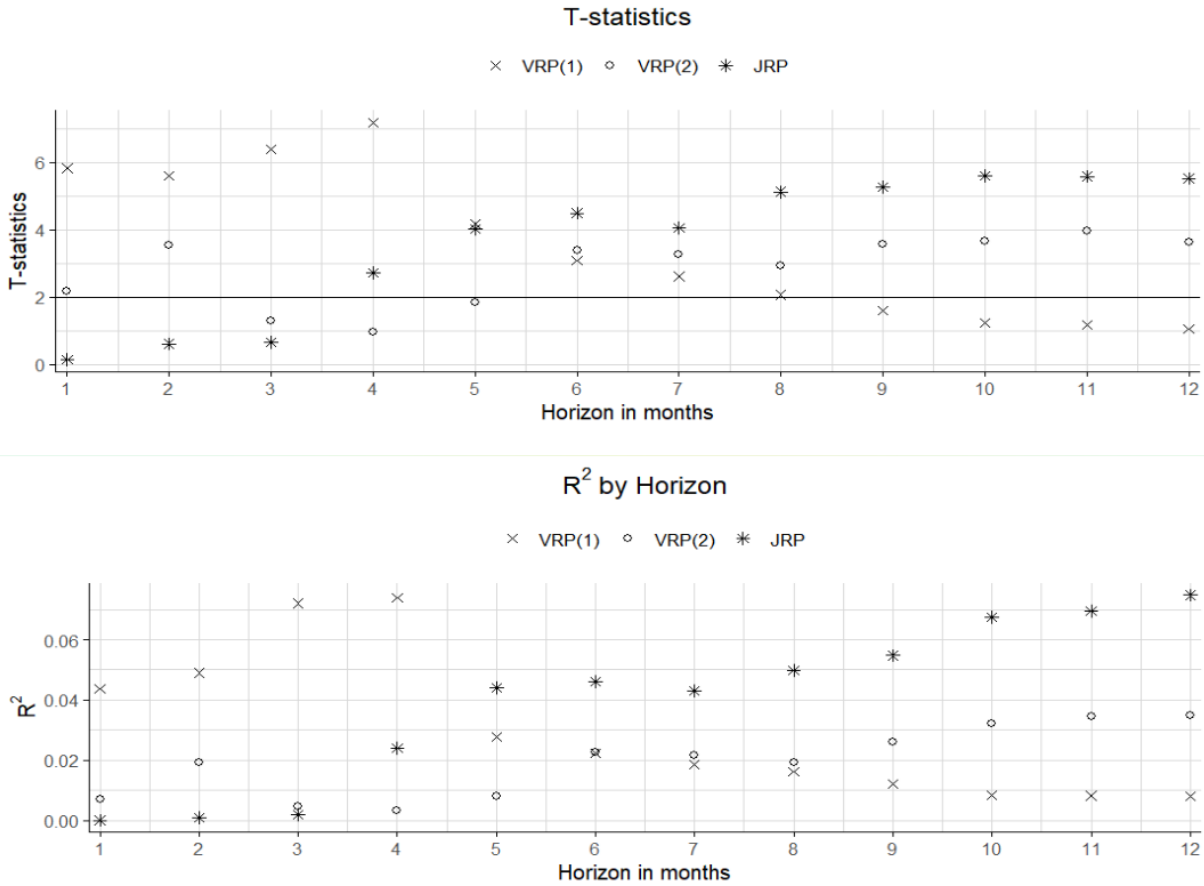

Figure 2: Return predictability results. Standard errors are computed using Newey-West heteroskedasticity and autocorrelation consistent (HAC) standard errors, with lag order set equal to the forecasting horizon. The one-sided 5% critical value equals 1.99 and is plotted as a solid line in the top panel.

D Residual Diagnostics

Table 9: Residual Diagnostics for Huber-Estimated HAR Models

Model	**Skewness**	**Kurtosis**	**JB**
HAR-TV (log)	-2.742	23.772	67,963.164***
HAR-MV (log)	-1.188	8.396	8,696.081***
HAR-EV (log)	-1.141	6.515	5,443.454***

Model	**ARCH(5)**	**ARCH(22)**	**Q(5)**	**Q(22)**	**Q2(5)**	**Q2(22)**
HAR-TV (log)	1,031.568***	1,129.700***	5,651.624***	14,693.190***	1,968.950***	4,013.598***
HAR-MV (log)	739.443***	755.574***	5,358.747***	12,120.090***	1,591.136***	2,048.034***
HAR-EV (log)	1,265.558***	1,316.512***	5,509.276***	14,121.400***	3,296.115***	5,394.039***

Note: The table reports diagnostic statistics of the residuals of the HAR-TV, HAR-MV and HAR-EV models in forecasting one-day-ahead log *RV*, estimated using Huber regression. The regressions are conducted over the entire sample from 3 January 2008 until 31 December 2018. JB depicts the test statistic of the Jarque-Bera test. ARCH(·) represents ARCH-LM test statistics. Q2(·) represents the Q-statistics of the Ljung-Box test for the squared residuals. *** denotes rejection of the null hypothesis at the 1% significance level.

E Comparative Results for Student's-t Distribution

E.1 Probability-Integral Transformation (PIT) Histograms

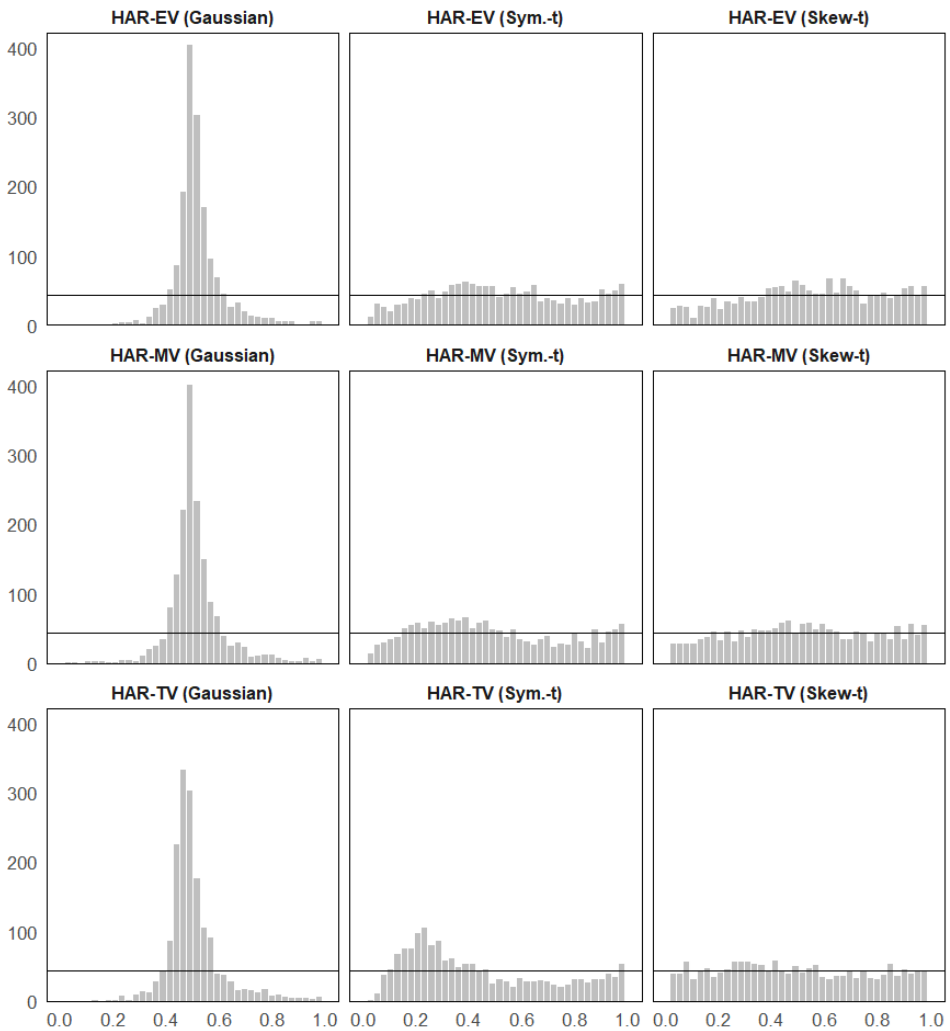

Figure 3: Probability-Integral Transformation (PIT) histograms for one-day-ahead RV density forecasts. The histograms are presented for the HAR model with Gaussian innovations, the HAR model with a GARCH(1,1) component and Student's-t innovations (Sym.-t), and the HAR model with a GARCH(1,1) component and skewed-t innovations (Skew-t).

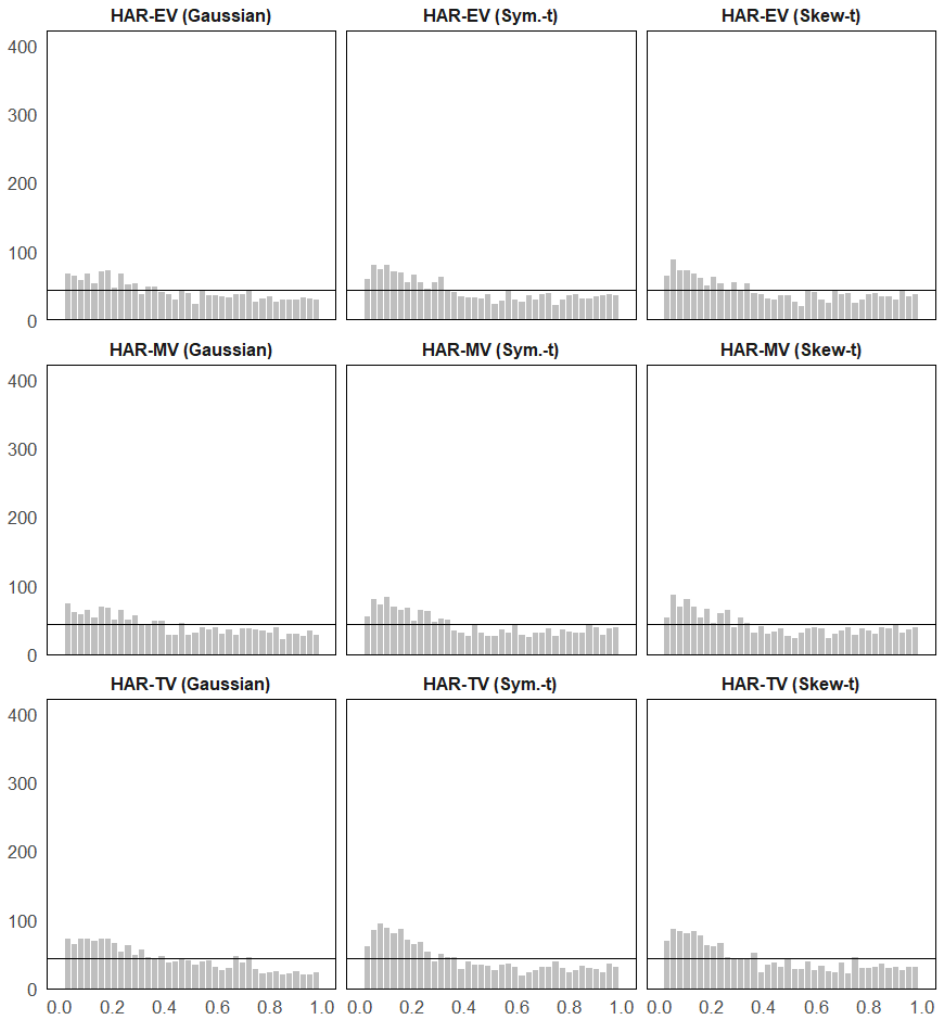

Figure 4: Probability-Integral Transformation (PIT) histograms for one-day-ahead log RV density forecasts. The histograms are presented for the HAR model with Gaussian innovations, the HAR model with a GARCH(1,1) component and Student's-t innovations (Sym.-t), and the HAR model with a GARCH(1,1) component and skewed-t innovations (Skew-t).

E.2 Amisano-Giacomini Test Results

Table 10: Amisano and Giacomini (AG)-test results comparing the predictive accuracy of density forecasts resulting from flexible residual specifications (GARCH with Student's-t innovations) to density forecasts resulting from Gaussian residuals specifications.

Model	RV			$\log(RV)$		
	$\overline{\text{LPSD}}$	t-stat	p-value	$\overline{\text{LPSD}}$	t-stat	p-value
HAR-TV	2.2611	2.291	0.0219	0.0785	4.694	0.0000
HAR-MV	2.5437	2.056	0.0398	0.0800	4.553	0.0000
HAR-EV	2.4633	2.218	0.0266	0.0708	4.483	0.0000

Table 11: Amisano and Giacomini (AG)-test results comparing the predictive accuracy of density forecasts resulting from a specification with a GARCH component and Skewed-t innovations to density forecasts resulting from a specification with a GARCH component and Student's-t innovations.

Model	RV			$\log(RV)$		
	$\overline{\text{LPSD}}$	t-stat	p-value	$\overline{\text{LPSD}}$	t-stat	p-value
HAR-TV	0.1355	11.429	0.0000	0.0100	3.408	0.0007
HAR-MV	0.0369	6.369	0.0000	0.0084	3.827	0.0001
HAR-EV	0.0083	1.839	0.0658	0.0071	3.641	0.0003

E.3 Discussion

We compare the predictive performance of density forecasts resulting from a specification with Gaussian innovations, and from a flexible specification including a GARCH component and student's-t or skewed-t innovations. A regular student's-t distribution is symmetric around its mean, but allows for heavier tails than a Gaussian distribution. Figure 3 shows that the incorporation of the student's-t distribution combined with a GARCH component leads to z_t being closer to uniformity than the Gaussian counterpart. Table 10 indicates a similar finding, revealing a statistically significant improvement in predictive performance. Allowing for fat-tailed innovations and time-varying volatility leads to clear accuracy gains in density forecasting.

The skewed-t distribution is a generalization of the student's-t distribution that explicitly allows for asymmetry within the left and right tails of the distribution. Although not clearly visible in the logarithmic specification of Figure 4, the probability-integral transformations of Figure 3 indicate that accounting for skewness in the distribution may further improve predictive ability. Table 11 confirms this finding, showing that explicitly allowing for an uneven placement of probability mass in the left and right tails of the distribution leads to a better representation of the uncertainty around a point forecast. With the exception of the standard HAR-EV specification, the predictive accuracy of the density forecasts is significantly improved at the 1% confidence level.

F Probability-Integral Transformations

F.1 Standard Forecasting Targets

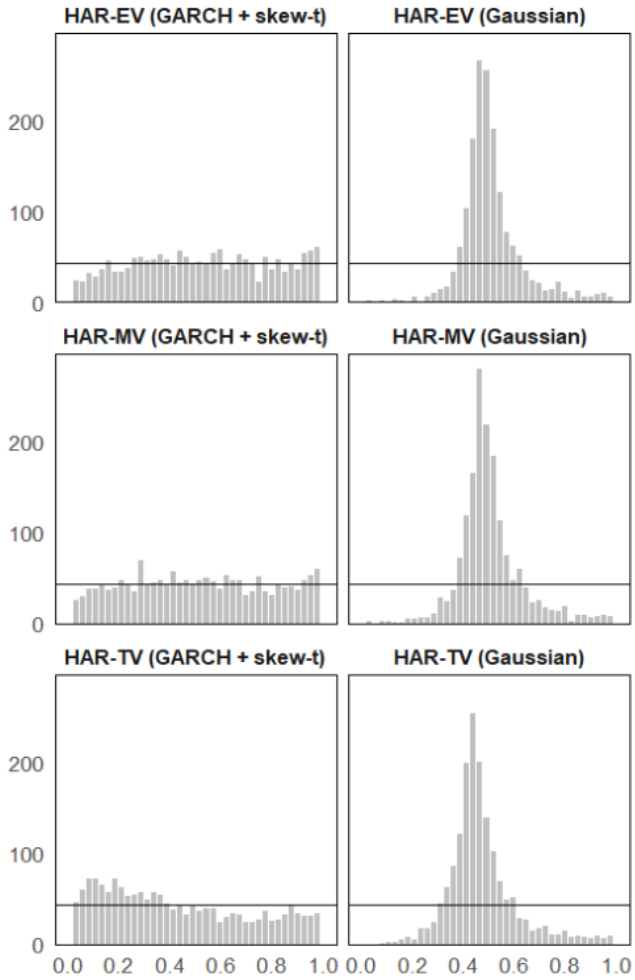

Figure 5: Probability-Integral Transformation (PIT) histograms for one-day-ahead *ORV* density forecasts. The histograms are presented for both the HAR model with Gaussian innovations and the HAR model with a GARCH(1,1) component and skewed-t innovations.

F.2 Logarithmic Forecasting Targets

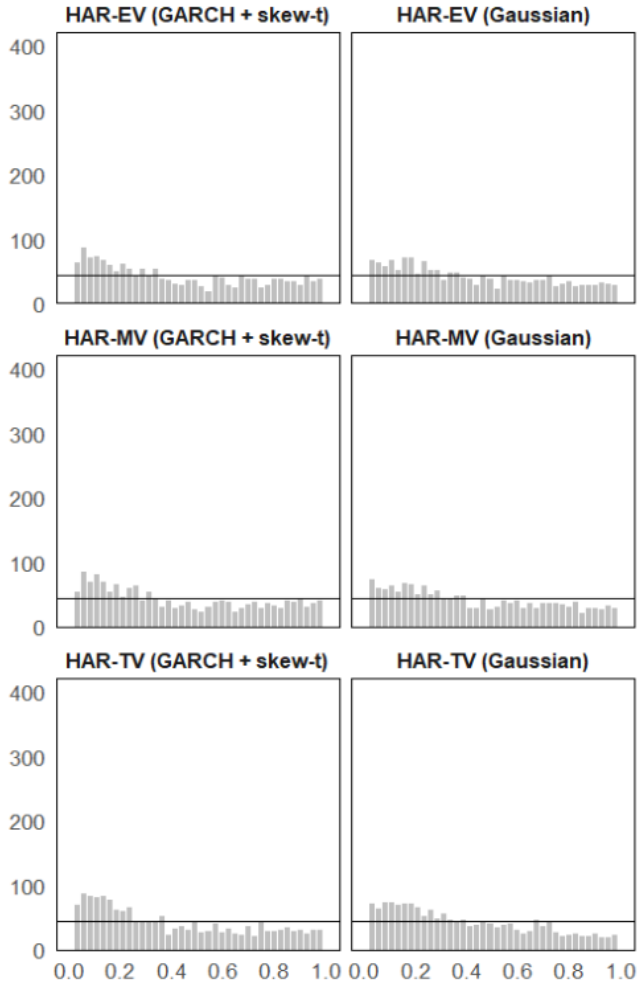

Figure 6: Probability-Integral Transformation (PIT) histograms for one-day-ahead log RV density forecasts. The histograms are presented for both the HAR model with Gaussian innovations and the HAR model with a GARCH(1,1) component and skewed-t innovations.

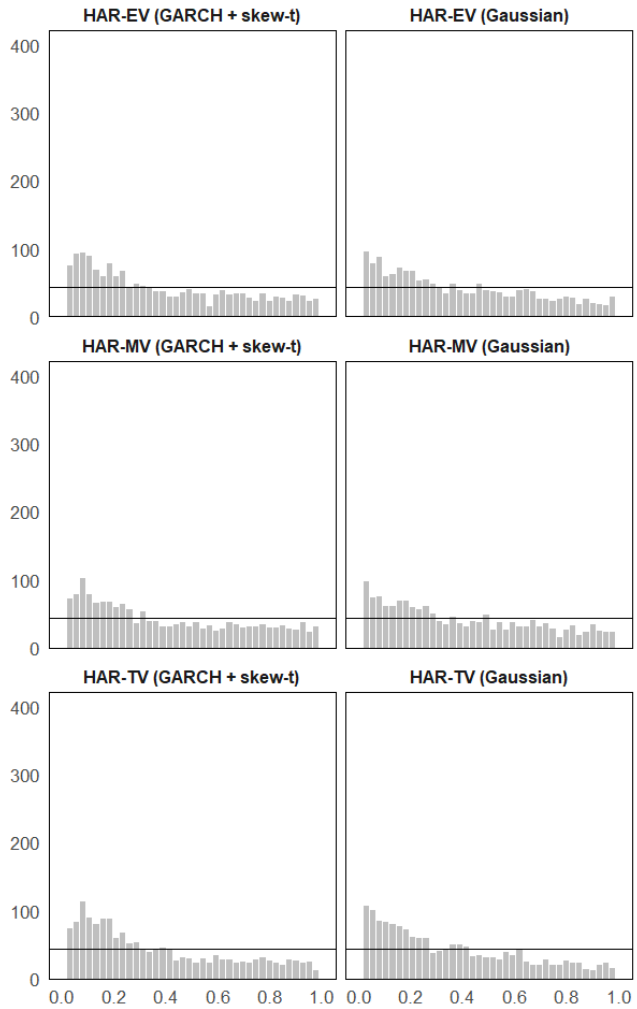

Figure 7: Probability-Integral Transformation (PIT) histograms for one-day-ahead log *ORV* density forecasts. The histograms are presented for both the HAR model with Gaussian innovations and the HAR model with a GARCH(1,1) component and skewed-t innovations.

G Flexible Residual Specification

Table 12: Amisano and Giacomini (AG)-test results comparing the predictive accuracy of *ORV* density forecasts resulting from flexible residual specifications (GARCH with skewed-t innovations) to density forecasts resulting from Gaussian residuals specifications.

Model	*ORV*			log(*ORV*)		
	$\overline{LPSD}$	t-stat	p-value	$\overline{LPSD}$	t-stat	p-value
HAR-TV	1.6929	2.402	0.0163	0.1709	6.406	0.0000
HAR-MV	2.0484	1.963	0.0497	0.1239	6.209	0.0000
HAR-EV	2.0440	2.009	0.0445	0.1141	6.578	0.0000